

Fine-Tuning Helps Against Blur and Hurts Against Noise: Transfer Strategies for Plant Disease Recognition under Field Capture

Egor Savchenko Zamir Safin Ilia Ponomarev Vadim Poponnikov
Innopolis University — Computer Vision 2026 — Project P02

1 Problem and question

PlantVillage is captured in a laboratory: one detached leaf, uniform background, even illumination. A pretrained ResNet-50 saturates on it, so the question worth asking is not whether transfer learning works but *which transfer strategy survives leaving the laboratory*. Kumar et al. [1] show that fine-tuning can distort pretrained features and lose to linear probing out of distribution. We ask whether that reversal appears in plant disease recognition, and if so, under which kind of degradation. The controlled variable is which parameters receive gradients; architecture, data, optimiser, schedule and seed budget are fixed.

2 Dataset

PlantVillage [2], Kaggle mirror, 38 classes over 14 crops. The distribution ships `train` and `val`; we build the test set by moving 5 % of each training class into `test` with seed 42, giving 41 275 / 10 861 / 2 169 images. Images are resized to 224×224 and normalised with ImageNet statistics; training adds flips and rotation up to 15°. Model selection uses validation macro F1 only.

The split is not grouped by leaf. PlantVillage holds several photographs of one physical leaf and Mohanty et al. kept them on one side of their splits; our file-level split does not. A 64-bit perceptual hash over all 43 444 images (`9_leakage_check.py`) finds one exact match and 51 test images, 2.35 %, within Hamming distance 5 of a training image, 44 of them in the same class. Removing all 51 moves clean accuracy by at most 10^{-4} in either direction, for every run of both conditions; one linear-probe error per seed falls on a flagged image and no fine-tuned error does. The contamination is real, measured, and does not carry the numbers. It is a lower bound: the hash catches only near-identical framing and is not invariant to rotation or mirroring.

3 Method

Both conditions start from the same `ResNet50.Weights.DEFAULT` checkpoint with a 38-way linear head. The **baseline** is a true linear probe: the backbone is frozen *and* every BatchNorm module is kept in evaluation mode during training, so the features are genuinely fixed (77 862 of 23 585 894 parameters trainable). The **second condition** fine-tunes all 23 585 894. Shared: AdamW, weight decay 10^{-4} , batch 64, mixed precision, `ReduceLROnPlateau` on validation macro F1, early stopping with patience 5.

Each condition’s learning rate was tuned separately on validation macro F1, since they solve different optimisation problems. The probe swept 10^{-4} to 10^{-1} for 0.9644 / 0.9762 / **0.9793** / 0.9771 / 0.9734, an interior optimum at 10^{-2} ; fine-tuning uses 10^{-4} , its sweep giving 3, 2 and 8 errors with fully overlapping intervals, so no ranking is claimed. Three seeds (0, 1, 42) per condition.

Table 1: Clean test set, seed 42, with 95 % Wilson intervals; the last row is the mean over three seeds.

	Linear probe	Fine-tuned
Trainable parameters	77 862	23 585 894
Epochs to early stop	36	16
Training time (min)	14.4	9.9
Accuracy	.9843 [.978,.989]	.9986 [.996,.9995]
Macro F1	.9804	.9976
Errors of 2 169	34	3
ECE	.0080	.0016
Accuracy, 3 seeds	.9822 ± .0021	.9988 ± .0003

Table 2: Accuracy under simulated robotic capture: mean over three seeds, [min, max] across seeds. Bold marks the winner; ranges never overlap except at low light s1, the one cell with no significant difference ($p = 0.11$).

Artifact	S	Linear probe	Fine-tuned
Motion blur	1	.899	.983
	2	.724 [.712,.746]	.837 [.820,.853]
	3	.524 [.505,.545]	.576 [.564,.585]
Low light + noise	1	.886 [.874,.893]	.890 [.876,.906]
	2	.776 [.755,.787]	.538 [.454,.610]
	3	.438 [.415,.459]	.136 [.131,.140]
Gaussian noise (control)	1	.890 [.877,.896]	.827 [.812,.857]
	2	.768 [.739,.787]	.485 [.415,.553]
	3	.417 [.395,.434]	.133 [.128,.137]
JPEG	1	.960	.996
	2	.838 [.833,.842]	.948 [.944,.953]
	3	.646 [.638,.652]	.761 [.756,.765]

Capture degradation. We model an autonomous field robot with artifacts applied to the test images in capture space at three severities, seeded per image: *motion blur*, a line kernel of 5/11/19 px at a random angle; *low light with sensor noise*, darkened by 0.50/0.30/0.18 with Poisson shot and Gaussian read noise at that level, then scaled back up like a camera raising gain: brightness returns, signal-to-noise ratio does not; and *JPEG* at quality 30/15/8. Because low light confounds darkening, noise and gain, a **control** adds Gaussian noise at unchanged brightness with σ matched to the output-referred noise of low light at mid-grey (0.066 / 0.105 / 0.221 against 0.0657 / 0.1048 / 0.2207). Neither model is retrained.

4 Results

The reversal is not a sampling artefact. Paired McNemar tests on identical inputs give $p = 1.05 \times 10^{-84}$ at low light severity 2 and $p = 6.9 \times 10^{-134}$ at severity 3 for the probe, against $p = 4.8 \times 10^{-40}$ for fine-tuning at motion blur severity 2; across seeds the best fine-tuned run at low light severity 2 (0.610) still falls below the worst probe run (0.755).

The control isolates the cause, not the mechanism. Gaussian noise at unchanged brightness reproduces the reversal and produces it earlier — the probe already wins at severity 1 ($p = 1.4 \times 10^{-20}$), where low light is a tie. Darkening and gain are not needed; noise is sufficient. How the noise acts

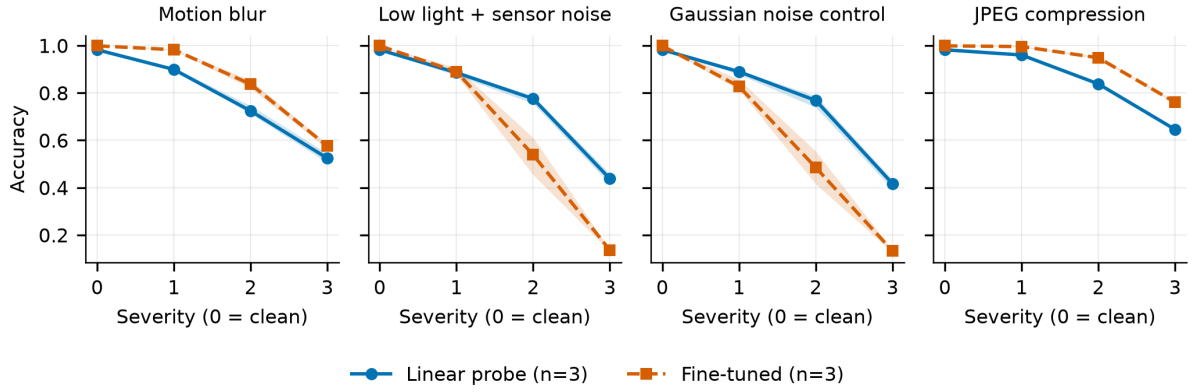


Figure 1: Degradation under the four artifacts; line is the mean over three seeds, band the min–max range. Fine-tuning dominates under blur and JPEG at every severity; under low light the curves cross between severity 1 and 2. The Gaussian control (third panel) reproduces that reversal without any darkening or gain, and one severity earlier: noise alone is sufficient to produce it.

is a separate question this control cannot answer. Inference cost is architectural: measured for both conditions on the same A100-PCIE-40 GB worker, 7.09 and 7.30 ms per image at batch 1, 0.355 ms for both at batch 64, 773.1 MB peak memory for both.

5 Interpretation

Our hypothesis, not tested here, is that fine-tuning on noise-free laboratory images tunes early filters to high-frequency texture statistics that only that capture regime produces. Blur and JPEG *remove or quantise* high frequencies, so the adapted mid-level features still carry the signal and fine-tuning wins by up to 11 points; noise *injects* random energy into exactly the band those filters occupy, with no generic representation left underneath. The frozen ImageNet backbone, trained on web photographs that already contain noise and compression, would keep a tolerance fine-tuning trades away; the direct test is band-limited noise of equal energy [4]. The *effect* matches Kumar et al. [1] (2 points gained in distribution, 7 lost out of it), whose proposed mechanism differs; we refine it: the reversal depends on the artifact, not on distribution shift in general.

6 Failure analysis

Clean data. Every error of the fine-tuned model, two or three per seed, confuses two diseases of the same crop; for the linear probe the same-crop share is 0.79 to 0.94 across seeds. Frequent pairs are Corn Cercospora/Gray leaf spot ↔ Northern Leaf Blight, Grape Esca → Black rot and Tomato Late blight → Early blight.

Under noise the fine-tuned model concentrates its errors. At Gaussian noise severity 2 it makes 969–1 268 errors across seeds, 61–76 % of them on one label, Tomato Septoria leaf spot, over 16–21 predicted classes. The probe makes 462–567 errors over 21–26 classes, its largest label taking 22–40 % (Target Spot in two seeds). Both drift toward speckled tomato diseases, so the label set supplies part of the pull and fine-tuning amplifies it. Septoria’s fine dark speckling resembles sensor noise: consistent with the hypothesis, not proof of it.

Calibration collapses under shift. On clean data fine-tuning is the better calibrated model (ECE 0.0007–0.0016 across seeds against 0.0078–0.0107). At low light severity 2 this inverts: 0.26–0.42 against 0.17–0.20. The model stops signalling exactly when it leaves its operating regime.

Limitations. The sensor model is an approximation: noise is applied to gamma-encoded sRGB, not to a linear signal, and severities were chosen by inspecting degraded test images. Training times come from different cluster workers. PlantVillage also carries a background shortcut — Noyan [3] reaches 49.0 % against 2.6 % for chance from eight background pixels. Next steps: a leaf-grouped split, corruption augmentation, and field images over the intersecting classes of PlantDoc.

7 Reproducibility

Repository: <https://github.com/EgorSavchenko-web/icv2026-plant-disease>.
 Python 3.12, torch 2.10.0, torchvision 0.25.0, CUDA 12.9, NVIDIA A100 under a ClearML agent; `requirements.txt` pins every dependency. The test split is committed with a manifest that `1_split_test.py` reproduces exactly, and release v1.1 carries all six checkpoints behind the tables plus the BatchNorm ablation, so every number can be verified with `5_inference.py` and `6_robustness_eval.py` without retraining. Corruptions are seeded per image; task ids, workers, hyperparameters and timings are in `results/run_provenance.json`.

8 Members’ contributions

Egor Savchenko — research question and experimental design; system architecture; training, inference, robustness and failure-analysis pipelines; verification and reproducibility. **Zamir Safin** — dataset selection and provenance; split construction and near-duplicate measurement; class balance; preprocessing and augmentation policy. **Iliia Ponomarev** — survey of existing plant-disease recognition solutions; backbone and transfer-strategy selection; positioning against published baselines. **Vadim Ponomnikov** — technical summary, presentation slides and the live demo.

References

- [1] A. Kumar et al. Fine-Tuning can Distort Pretrained Features and Underperform Out-of-Distribution. *ICLR*, 2022.
- [2] S. P. Mohanty et al. Using Deep Learning for Image-Based Plant Disease Detection. *Front. Plant Sci.*, 2016.
- [3] M. A. Noyan. Uncovering bias in the PlantVillage dataset. arXiv:2206.04374, 2022.
- [4] D. Yin et al. A Fourier Perspective on Model Robustness in Computer Vision. *NeurIPS*, 2019.